

Sodium-ion cells for stationary storage: cost and cycle life in controlled testing

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Abstract

We tested 48 prismatic sodium-ion cells (layered-oxide cathode, hard-carbon anode) under a stationary-storage duty cycle of one full cycle per day at 25 °C. Cells reached 80 percent of initial capacity after 4,200 cycles on average. At a production volume of 20 GWh a year we estimate a 2026 pack cost of 61 USD per kWh, 18 percent below LFP packs at 74 USD per kWh.

1. Method

Cells were cycled at 0.5C charge and discharge between 1.5 V and 4.0 V in a climate chamber held at 25 °C. Capacity was checked every 100 cycles. The study was funded by the cell maker Corvane Cells, which supplied all 48 cells.

2. Results

Mean cycle life to 80 percent capacity was 4,200 cycles (range 3,850 to 4,610). Round-trip efficiency was 92 percent. Capacity loss was linear after the first 200 cycles.

3. Cost

Bill of materials at 20 GWh a year gives a 2026 pack cost of 61 USD per kWh against 74 USD per kWh for LFP. The advantage comes from sodium carbonate and aluminium current collectors on both sides.

4. Limitations

All tests ran at a constant 25 °C. We did not test partial-state-of-charge cycling or cold climates.

Table 1. Cycle life by cell batch

Batch	Cells	Cycles to 80%
A	16	4180
B	16	4610
C	16	3850